

1 3

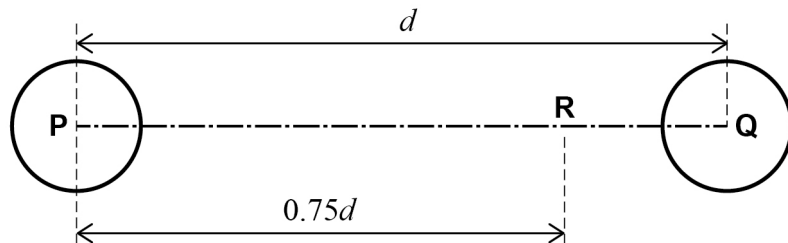
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The diagram shows two planets, **P** and **Q**.

The distance between the centres of the planets is d .

The resultant gravitational field strength at **R** is zero.

The distance **PR** is $0.75d$.



What is $\frac{\text{mass of P}}{\text{mass of Q}}$?

[1 mark]

A $\frac{1}{9}$



B $\frac{1}{3}$



C 3



D 9



1

4

The gravitational field strength at the surface of a uniform solid sphere is g .
The radius of the sphere is R .

What is the density of the sphere?

[1 mark]

A $\frac{g}{4\pi G}$



B $\frac{3g}{4\pi GR}$



C $\frac{4g}{3\pi GR}$



D $\frac{4\pi GR}{3g}$



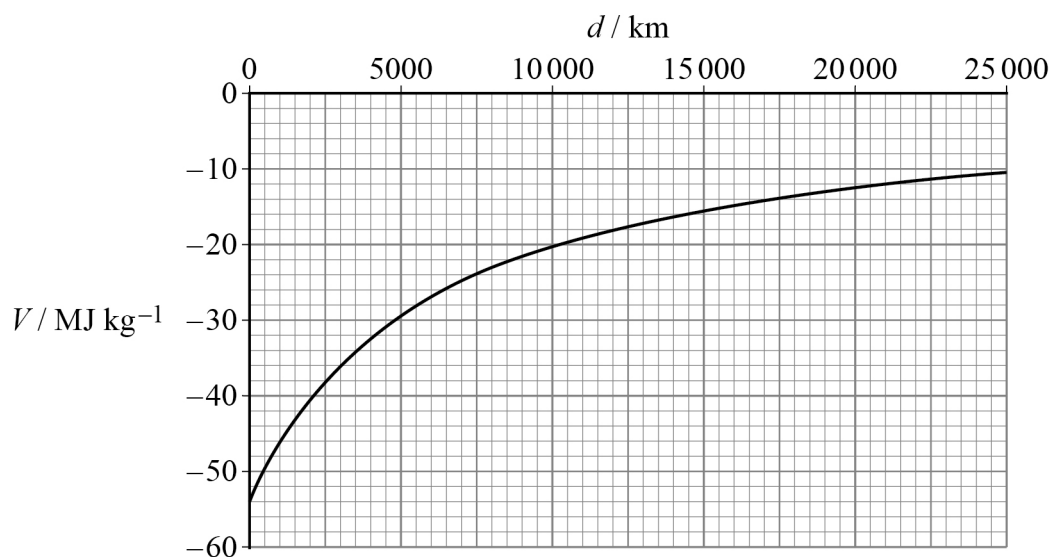
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Do not write Do not write

1 5

The graph shows the variation of the gravitational potential V with distance d from the surface of Venus.



What is the gravitational field strength when $d = 10\,000\text{ km}$?

[1 mark]

- A** -2.0 N kg^{-1} ☐
- B** -1.3 N kg^{-1} ☐
- C** $+1.3\text{ N kg}^{-1}$ ☐
- D** $+2.0\text{ N kg}^{-1}$ ☐



Don't write

0 4

This question is about the gravitational fields of the Earth and its satellites.

0 4 . 1

The Moon orbits the Earth with a time period of 27.3 days.

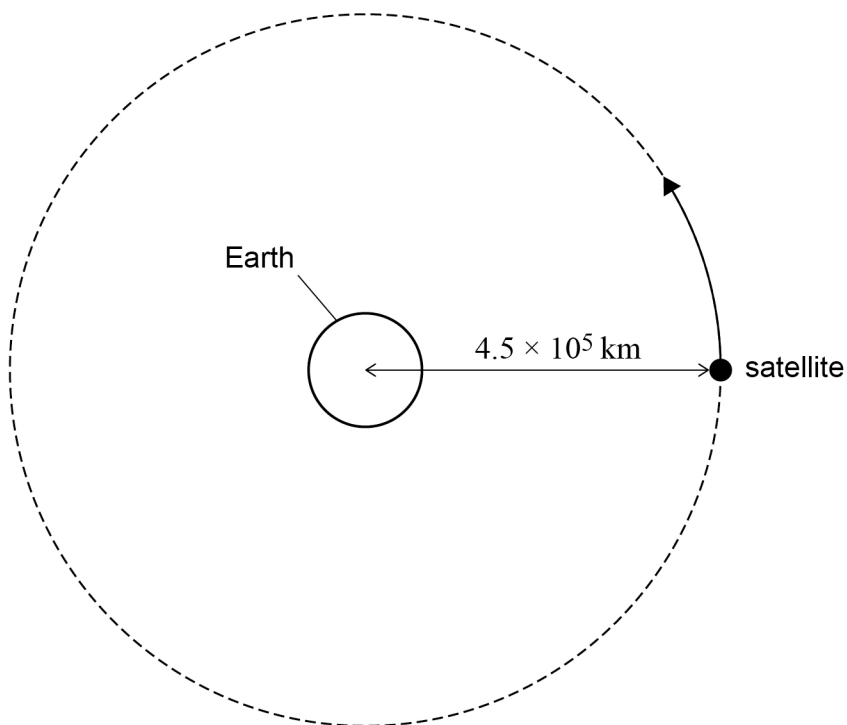
Show that the angular speed of the Moon is approximately $2.7 \times 10^{-6} \text{ rad s}^{-1}$.

[2 marks]

0 4 . 2

Figure 7 shows an artificial satellite in a circular orbit of radius $4.5 \times 10^5 \text{ km}$ from the centre of the Earth.

Figure 7



Calculate the gravitational field strength of the Earth at this radius.

[2 marks]

gravitational field strength = _____ N kg^{-1}

0 4 . 3

The satellite and the Moon orbit the Earth at the same angular speed.

Calculate the centripetal acceleration of the satellite.

[2 marks]

centripetal acceleration = _____ m s^{-2}

Question 4 continues on the next page

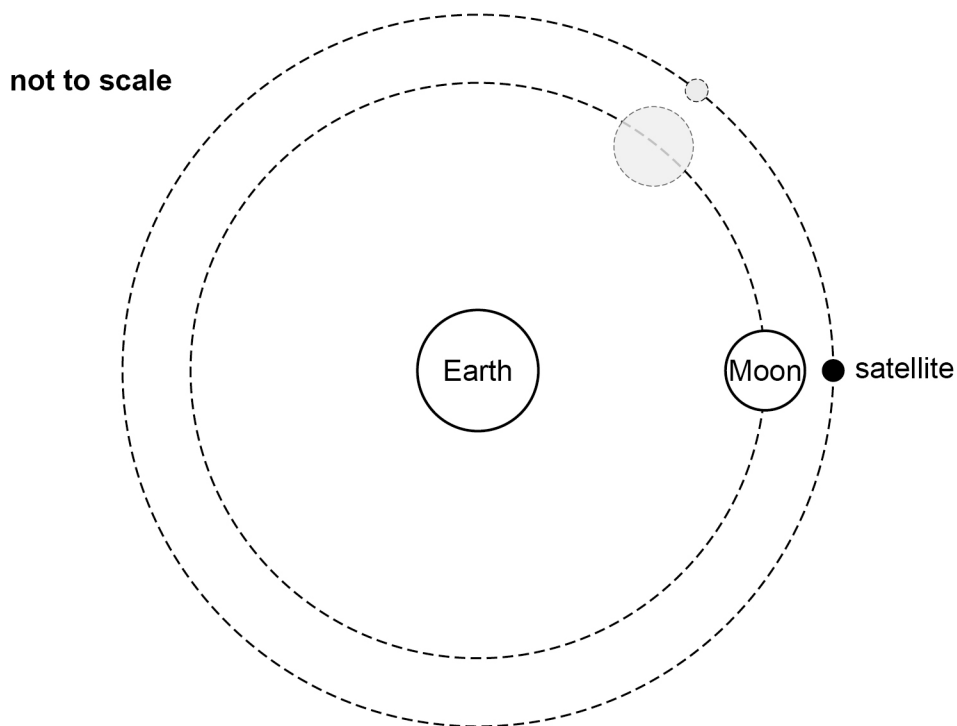
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0 4 . 4

Figure 8 shows two positions of the satellite and the Moon as they orbit the Earth. The orbital radius of the satellite is greater than the orbital radius of the Moon but their orbital periods are the same.

Figure 8



Explain how this satellite orbit is possible.

[2 marks]



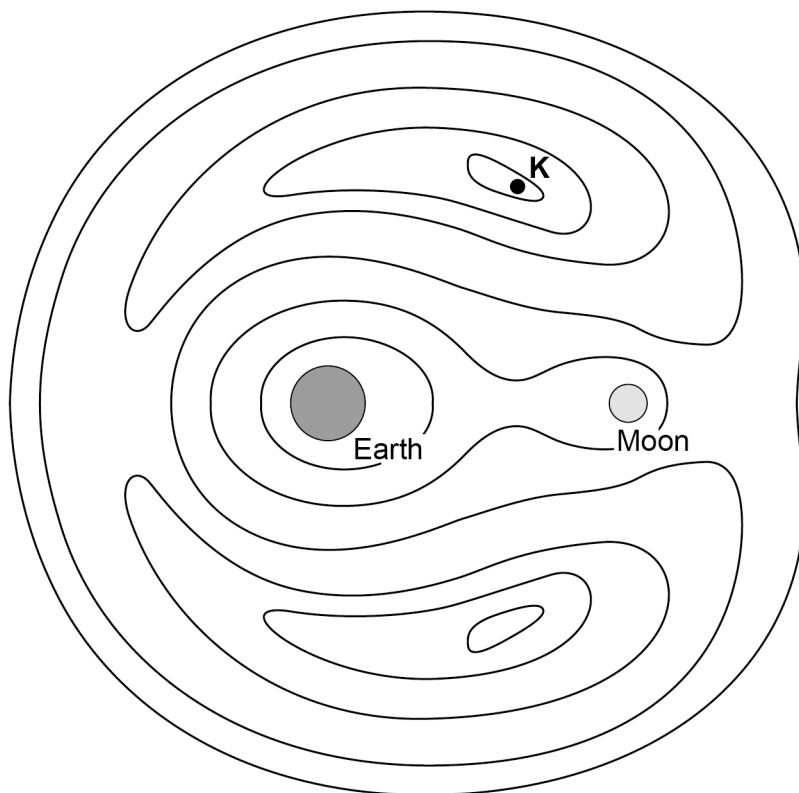
0 4 . 5

Figure 9 shows some resultant gravitational equipotential lines for the Earth and Moon system.

Draw, on **Figure 9**, **one** gravitational field line from the Moon to **K**.

[1 mark]

Figure 9



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**



Don't write

1**2**

A planet has a density of $1.9 \times 10^3 \text{ kg m}^{-3}$ and a diameter of $5.2 \times 10^3 \text{ km}$.

What is the gravitational field strength at the surface of the planet?

[1 mark]

A $1.4 \times 10^{-3} \text{ N kg}^{-1}$

☐

B $2.8 \times 10^{-3} \text{ N kg}^{-1}$

☐

C 1.4 N kg^{-1}

☐

D 2.8 N kg^{-1}

☐

1 3

A satellite is moved from a higher orbit to a lower orbit.

What energy changes occur?

[1 mark]

	Kinetic energy	Gravitational potential energy	Total energy	
A	decreases	decreases	increases	☐
B	decreases	increases	decreases	☐
C	increases	decreases	increases	☐
D	increases	decreases	decreases	☐



Don't waste

The electron in a hydrogen atom orbits the proton at a constant distance of 5.3×10^{-11} m.

What is the work done W to ionise this hydrogen atom?
Ignore any kinetic energy of the electron.

[1 mark]

A
$$\frac{2 \times 1.60 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times 5.3 \times 10^{-11}}$$
 ☐

B
$$\frac{2 \times 1.60 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times (5.3 \times 10^{-11})^2}$$
 ☐

C
$$\frac{(1.60 \times 10^{-19})^2}{4\pi \times 8.85 \times 10^{-12} \times 5.3 \times 10^{-11}}$$
 ☐

D
$$\frac{(1.60 \times 10^{-19})^2}{4\pi \times 8.85 \times 10^{-12} \times (5.3 \times 10^{-11})^2}$$
 ☐

0 1

One of the moons of Jupiter is named Io.

The mass of Io is $8.93 \times 10^{22} \text{ kg}$.

The gravitational potential due to Io at its surface is $-3.27 \times 10^6 \text{ J kg}^{-1}$.

0 1

1

State what is meant by a gravitational potential of $-3.27 \times 10^6 \text{ J kg}^{-1}$ at the surface of Io.

[2 marks]

0 1

2

Show that the radius of Io is approximately $1.8 \times 10^6 \text{ m}$.

[1 mark]



A volcanic rock of mass 3.00 kg is ejected vertically upwards from the surface of Io with an initial velocity of 760 m s^{-1} .

The rock reaches a maximum height h above the surface of Io.

0 1 . 3

Show that the rock's gravitational potential energy at height h is approximately -9 MJ .
[3 marks]

0 1 . 4

Calculate h .

[2 marks]

$h =$ _____ m

0 1 . 5

Another rock of greater mass is ejected vertically upwards from the surface with the same initial velocity.

State and explain how the height reached by the rock of greater mass compares to h .

[2 marks]



Do not write Do not write

1**2**

Two satellites **X** and **Y** have the same mass and orbit the same planet.

X has total energy of magnitude E and orbital radius r .

Y has orbital radius $2r$.

What is the magnitude of the total energy of **Y**?

[1 mark]

A $\frac{E}{4}$



B $\frac{E}{2}$



C E



D $2E$



Turn over ►



Do not write

0 5

Lunar Prospector was a space probe that orbited the Moon.
Table 1 shows data about the Moon and Lunar Prospector.

Table 1

Moon	mass / kg	7.348×10^{22}
	radius / m	1.737×10^6
Lunar Prospector	mass / kg	1.263×10^2

Lunar Prospector had an initial orbital radius of 1.84×10^6 m.

0 5

1

Calculate the acceleration of Lunar Prospector in this orbit.

[2 marks]

acceleration = _____ m s^{-2}

0 5

2

Calculate the orbital period of Lunar Prospector in this orbit.

[2 marks]

orbital period = _____ s



Lunar Prospector was moved to a different orbit where its gravitational potential energy (GPE) was $-3.47 \times 10^8 \text{ J}$.

0 5 . 3

Calculate the height above the Moon's surface of this different orbit.

[3 marks]

height = _____ m

0 5 . 4

The speed of Lunar Prospector in this orbit was 1.66 km s^{-1} .

Lunar Prospector left orbit and crashed onto the surface of the Moon.

Calculate the kinetic energy of Lunar Prospector when it hit the surface of the Moon. Assume that the contribution to the kinetic energy from Lunar Prospector's engines was negligible.

[3 marks]

kinetic energy = _____ J

Question 5 continues on the next page

Turn over ►



0 5 . 5

There are variations in the density of the material of the surface of the Moon. Measurements of Lunar Prospector in orbit were used to detect these variations.

At one point in its orbit, Lunar Prospector passed over a region where the density of the surface was greater than average.

Suggest how this affected the centripetal acceleration of Lunar Prospector.

[2 marks]

Don't write

A satellite orbiting the Earth has an orbital radius r and orbital speed v .

Which is correct?

[1 mark]

A $r \propto v^2$

☐

B $r^2 \propto v$

☐

C $r \propto \frac{1}{v}$

☐

D $r \propto \frac{1}{v^2}$

☐

Do not write Do not write

2**1**

A planet has a mass half that of the Earth and a density half that of the Earth.

What is the gravitational field strength on the surface of the planet?

[1 mark]

A 2.5 N kg^{-1}

☐

B 3.5 N kg^{-1}

☐

C 4.9 N kg^{-1}

☐

D 9.8 N kg^{-1}

☐

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

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2 2 1 X P H 0 3

0 2 . 1

Show that the total energy E of a satellite orbiting the Moon is given by

$$E = -\frac{GMm}{2r}$$

where m is the mass of the satellite, M is the mass of the Moon and r is the orbital radius of the satellite.

[3 marks]

0 2 . 2

Explain the significance of the negative sign in the equation in Question **02.1**.

[2 marks]

Question 2 continues on the next page



The satellite has a mass of 1.0×10^3 kg and orbits the Moon in a circular orbit.
The mass of the Moon is 7.3×10^{22} kg.

0 2 . 3

Determine the orbital speed of the satellite when the orbital radius is 1.95×10^3 km.

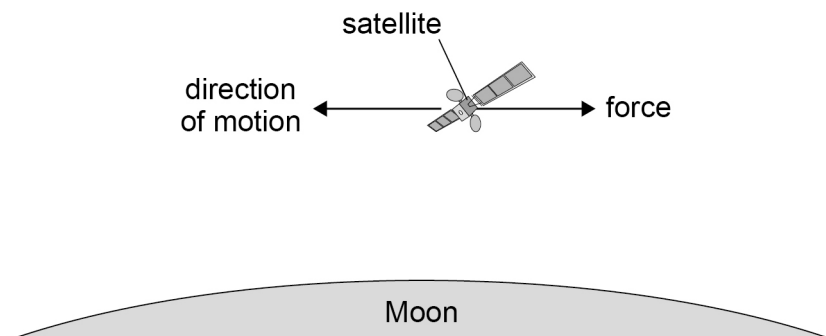
[3 marks]

orbital speed = _____ m s^{-1}



The satellite's thrusters exert a force in the direction opposite to its direction of motion, as shown in **Figure 5**. This causes the radius of the orbit to decrease.

Figure 5



0 2 . 4

Explain, with reference to energy, why the use of the thrusters reduces the radius of the orbit.

[2 marks]

0 2 . 5

Calculate the work done by the thrusters in reducing the radius of the satellite's orbit. Assume that the final orbit of the satellite is circular.

initial radius of the orbit = 1.95×10^3 km

final radius of the orbit = 1.85×10^3 km

[2 marks]

work done = _____ J



Do not write Do not write

1**2**

The average density of the Earth is 1.4 times greater than the average density of planet **X**.
The radius of the Earth is 1.9 times greater than the radius of **X**.

What is the gravitational field strength on the surface of **X**?

[1 mark]

A 3.7 N kg^{-1}

☐

B 7.2 N kg^{-1}

☐

C 13 N kg^{-1}

☐

D 26 N kg^{-1}

☐

Do not write

Comets are balls of rock and ice.
Figure 12 shows a comet in an elliptical orbit around the Sun. The comet is closest to the Sun at point **P**.

Figure 12

not to
scale

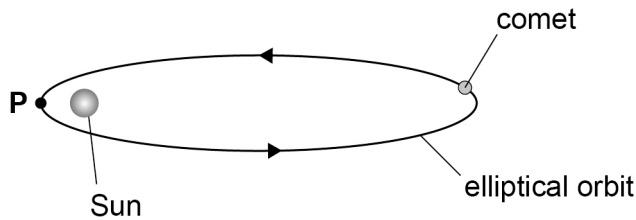


Table 1 shows data for the comet at **P**.

Table 1

distance to Sun	$8.758 \times 10^{10} \text{ m}$
speed	$5.457 \times 10^4 \text{ m s}^{-1}$
mass of comet	$2.242 \times 10^{14} \text{ kg}$

Show that the gravitational force of the Sun on the comet at **P** is approximately $3.9 \times 10^{12} \text{ N}$.

[1 mark]



07.2

Explain how the gravitational force in Question **07.1** and the data in **Table 1** show that the comet is **not** in a circular orbit around the Sun.

[2 marks]

07.3

The orbital period T of an object orbiting the Sun at a mean orbital radius R is approximately given by:

$$T^2 = kR^3$$

where k is a constant.

The mean orbital radius of the comet is $18R_E$ where R_E is the mean orbital radius of the Earth.

Estimate, in years, the orbital period T of the comet.

[3 marks]

$T =$ _____ years



0 7 . 4

The total energy of the comet is the sum of its kinetic energy and its gravitational potential energy.

Show that the total energy of the comet at **P** is approximately -6×10^{21} J.

[2 marks]

0 7 . 5

Calculations for a different comet show that it has a total energy of $+6 \times 10^{21}$ J at its closest point to the Sun.

A student suggests that this comet will **not** remain in orbit around the Sun.

Discuss whether this suggestion is correct.

[2 marks]

10

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

The mass of the Earth is 81 times the mass of the Moon.

A spaceship travels from the Earth to the Moon. The gravitational potential is a maximum at a point **P**.

At **P**, what is $\frac{\text{distance from centre of Earth}}{\text{distance from centre of Moon}}$?

[1 mark]

A 0.1

☐

B 1

☐

C 9

☐

D 81

☐

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1	1
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The gravitational field strength on the surface of a planet is g .
The diameter of the planet is d .

What is the mean density of the planet?

[1 mark]

A $\frac{g}{4\pi dG}$



B $\frac{g}{2\pi dG}$



C $\frac{3g}{4\pi dG}$



D $\frac{3g}{2\pi dG}$

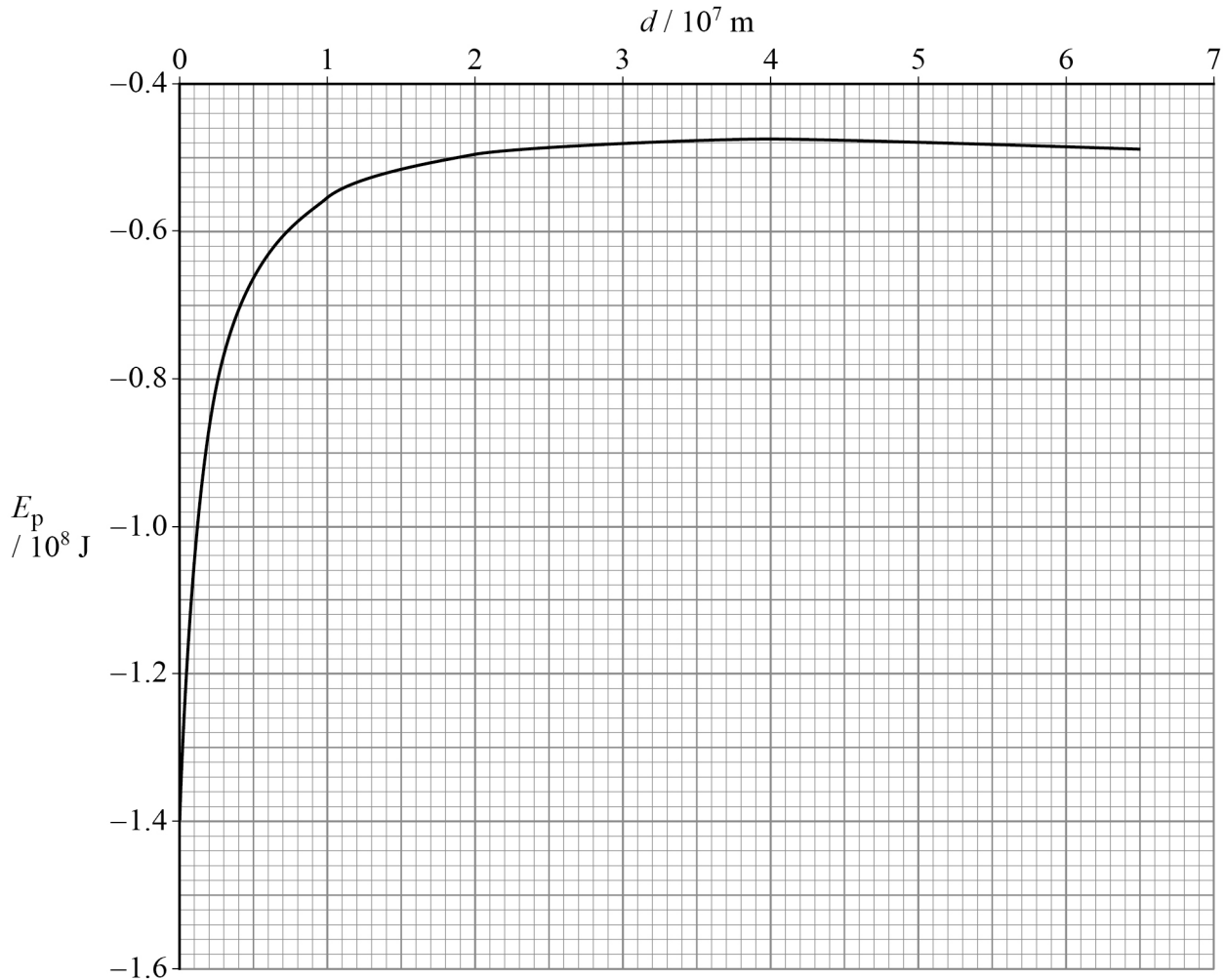


0 3

An asteroid hits the surface of the Moon and causes pieces of Moon rock to travel in many different directions. One rock travels along a line joining the centres of the Moon and the Earth.

Figure 7 shows the variation of gravitational potential energy E_p of the rock with the height d above the Moon's surface in the direction of the Earth.

Figure 7



0 3

1

Explain why the values of E_p in **Figure 7** are negative.

[2 marks]



0 3 . 2

The value of E_p is a maximum where d is approximately 4×10^7 m, as shown in **Figure 7**.

Explain, without calculation, why there is a maximum value of E_p as the rock travels from the Moon to the Earth.

[2 marks]

The mass of the rock is 37 kg.

0 3 . 3

Calculate, using **Figure 7**, the gravitational field strength where $d = 1.0 \times 10^7$ m.

[3 marks]

gravitational field strength = _____ N kg⁻¹

Question 3 continues on the next page

Turn over ►



0 3 . 4

The value of E_p at the surface of the Moon is $-1.4 \times 10^8 \text{ J}$.

The rock leaves the surface of the Moon with a speed v .

Determine, using data from **Figure 7**, the minimum value of v needed for the rock to reach the Earth.

[3 marks]

$v =$ _____ m s^{-1}

10



Don't write

A satellite of mass m is in a circular orbit around a planet of mass M .

The orbital radius R of the satellite and the period T of its orbit are related by the equation

$$T^2 = kR^3$$

What is k ?

[1 mark]

A $\frac{2\pi}{GMm}$



B $\frac{4\pi^2}{GMm}$



C $\frac{2\pi}{GM}$



D $\frac{4\pi^2}{GM}$



Don't write

1 **9** A satellite is in a geostationary orbit.

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What is the height of the satellite above the surface of the Earth?

[1 mark]

A $5.1 \times 10^6 \text{ m}$ ☐

B $3.6 \times 10^7 \text{ m}$ ☐

C $4.2 \times 10^7 \text{ m}$ ☐

D $2.2 \times 10^9 \text{ m}$ ☐

0	2
---	---

An unpowered spacecraft **B** orbits the Sun.

The orbit is not circular.

Assume that the only gravitational force on **B** is due to the Sun.

The following data are about **B**:

mass of **B** = 1600 kg

smallest distance between **B** and the centre of the Sun = 4.20×10^{10} m

largest distance between **B** and the centre of the Sun = 1.37×10^{11} m.

0	2	.	1
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Calculate the maximum gravitational force of the Sun on **B**.

[2 marks]

maximum gravitational force = _____ N

0	2	.	2
---	---	---	---

The total orbital energy of a satellite is the sum of its kinetic energy and its gravitational potential energy.

The total energy of **B** is -1.19×10^{12} J.

Calculate the minimum speed of **B**.

[3 marks]

minimum speed = _____ m s^{-1}



An unpowered spacecraft **C** is in a circular orbit of the Sun.

C has an orbital period of 1.45×10^7 s.

0 2 . 3

Calculate the orbital radius of **C**.

[3 marks]

orbital radius = _____ m

0 2 . 4

Explain which spacecraft, **B** or **C**, is moving on an equipotential.

[1 mark]

Turn over for the next question



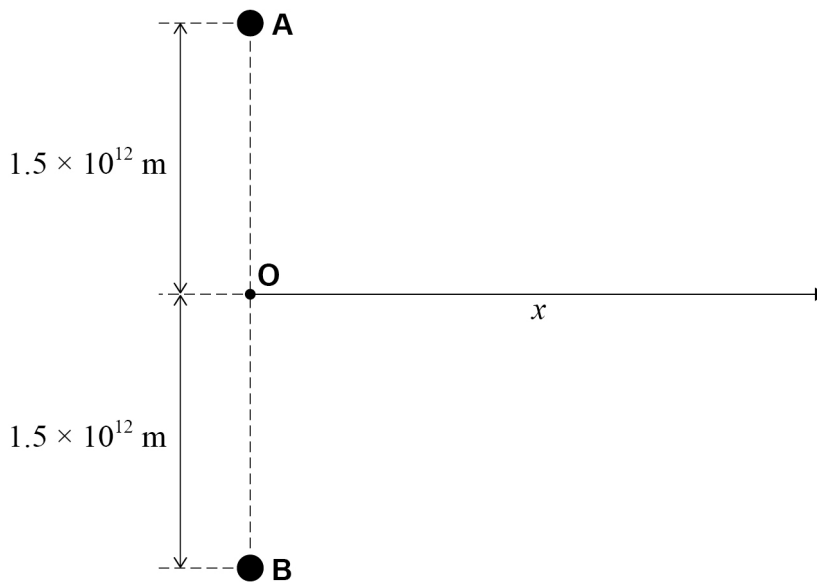
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0 4 . 1

Define gravitational potential at a point.

[2 marks]

Figure 5 shows a binary star system consisting of two stars **A** and **B**. **A** and **B** each have the same mass and their centres are separated by a distance of 3.0×10^{12} m. **O** is the point midway between **A** and **B**.

Figure 5

x is the distance from **O** along a line perpendicular to **AB**.

V_T is the total gravitational potential due to both **A** and **B**.

0 4 . 2

When $x = 0$ the value of V_T is $-0.711 \text{ GJ kg}^{-1}$.

Calculate the mass of **A**.

[2 marks]

mass of **A** = _____ kg

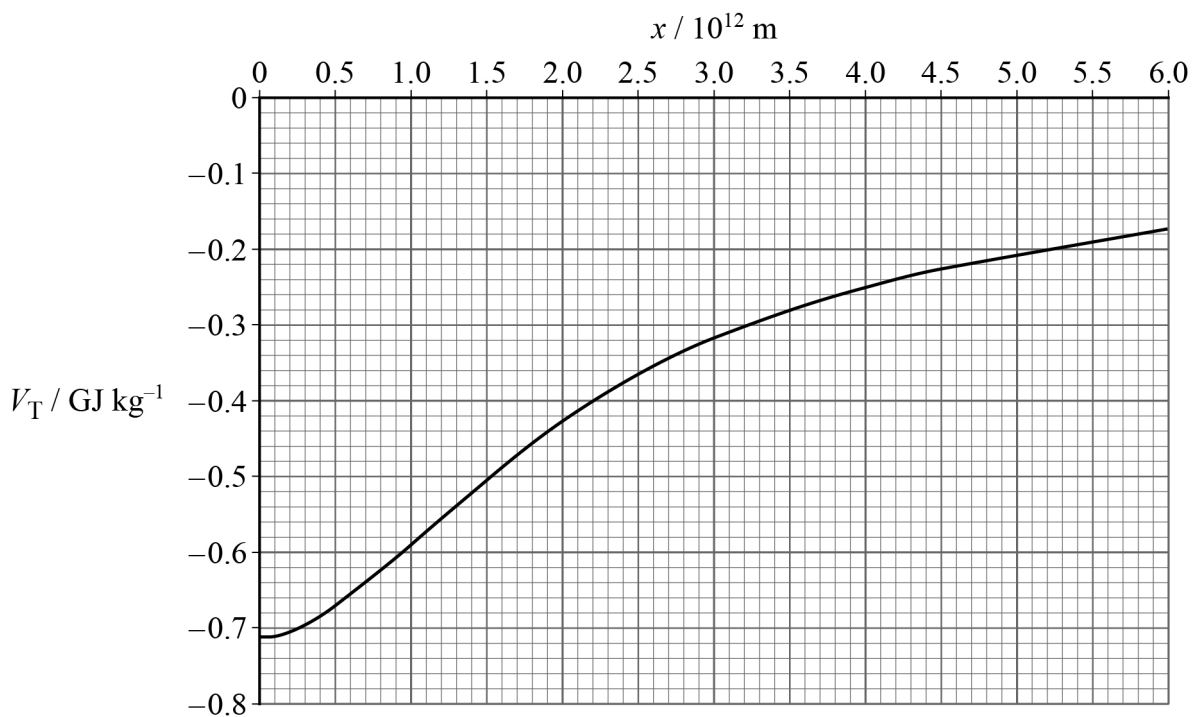
Question 4 continues on the next page

Turn over ►



Figure 6 shows the variation of V_T with x .

Figure 6



The magnitude of the resultant gravitational field strength due to **A** and **B** is g_T .

State the value of x at which g_T has its maximum value on **Figure 6**.
Go on to estimate this value of g_T .

Annotate **Figure 6** to support your calculation.

[3 marks]

$x =$ _____ m

maximum value of $g_T =$ _____ N kg^{-1}



0 4 . 4

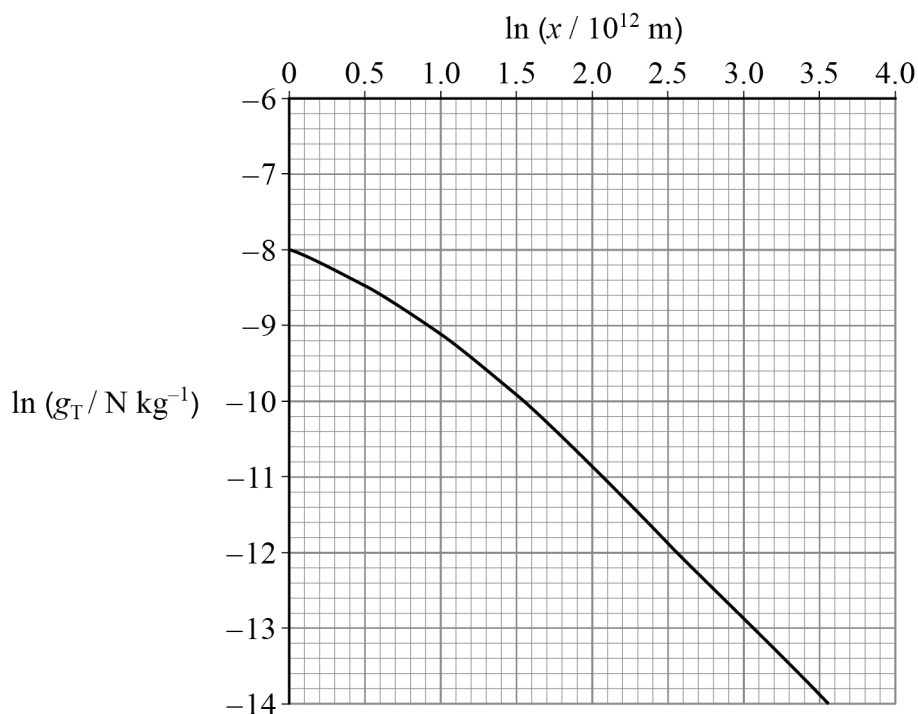
Describe the variation of g_T with x shown in **Figure 6**.

[2 marks]

0 4 . 5

Figure 7 shows the variation of $\ln g_T$ with $\ln x$ for larger values of x than those shown in **Figure 6**.

Figure 7



The variation of g_T with x follows an inverse-square law when x is greater than a minimum value $x_{\min}$.

Determine $x_{\min}$.

Annotate **Figure 7** to support your answer.

[3 marks]

$x_{\min} =$ _____ m

Do not write

1	0
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A satellite orbits the Earth at a constant orbital speed of v .
The radius of the orbit is R .

The satellite is moved to a new orbit of the Earth where it has a constant orbital speed of $2v$.

What is the radius of the new orbit?

[1 mark]

A $\frac{R}{4}$



B $\frac{R}{2}$



C $2R$



D $4R$

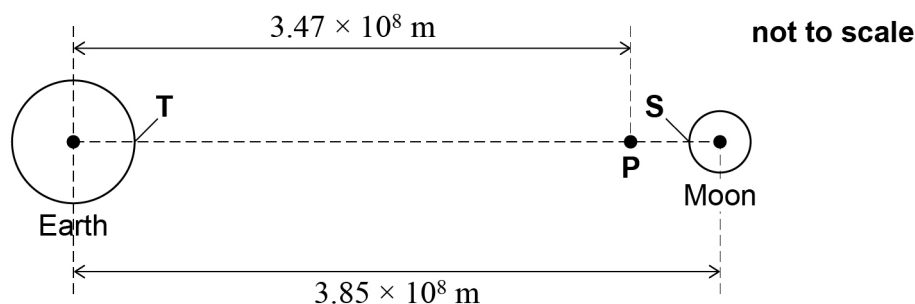


0 5

Figure 8 shows the Earth, the Moon and points **P**, **S** and **T**.
P is the point where the resultant gravitational field strength is zero.
S is the point on the surface of the Moon closest to the Earth.
T is the point on the surface of the Earth closest to the Moon.

P is a distance of 3.47×10^8 m from the centre of the Earth.

Figure 8



The distance between the centre of the Earth and the centre of the Moon, as shown in **Figure 8**, is 3.85×10^8 m.

0 5 . 1

Explain why the resultant gravitational field strength is zero at **P**.

[1 mark]

0 5 . 2

Determine the mass of the Moon.

[2 marks]

mass of Moon = _____ kg

Question 5 continues on the next page

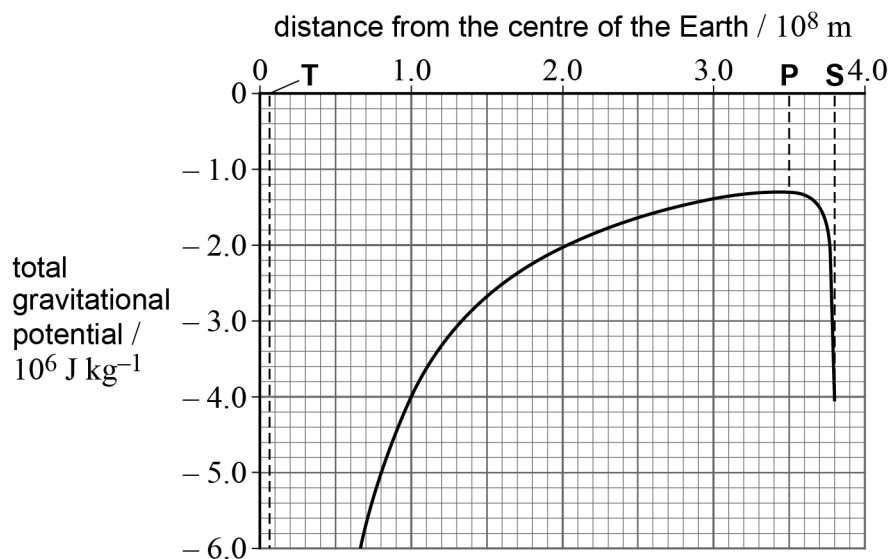
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Figure 9 shows the variation of the total gravitational potential with distance from the centre of the Earth towards the centre of the Moon.

P, **S** and **T** are shown on **Figure 9**.

Figure 9



0 5 . 3

Explain why the total gravitational potential at **P** is **not** zero.

[1 mark]



0 5 . 4

Use **Figure 9** to calculate the resultant gravitational field strength at a distance of 2.0×10^8 m from the centre of the Earth.

[3 marks]

resultant gravitational field strength = _____ N kg^{-1}

0 5 . 5

The total gravitational potential at **T** is $-6.26 \times 10^7 \text{ J kg}^{-1}$.

Determine the minimum energy required to move an object of mass 950 kg from the surface of the Earth to the surface of the Moon.
Ignore any effects due to the Earth's atmosphere.

[2 marks]

energy required = _____ J



Do not write

A satellite travels at a constant speed in a circular orbit around the Earth.

Which statement is correct?

[1 mark]

A The momentum of the satellite is constant.

☐

B The satellite accelerates towards the centre of the Earth.

☐

C The gravitational force on the satellite is negligible.

☐

D The satellite does not exert any force on the Earth.

☐

Do not write

1 3

Two satellites **X** and **Y** are in circular orbits around the Earth.

X and **Y** have the same mass.

The orbital radius of **X** is less than the orbital radius of **Y**.

X has an orbital period T_X and a total energy E_X .

Y has an orbital period T_Y and a total energy E_Y .

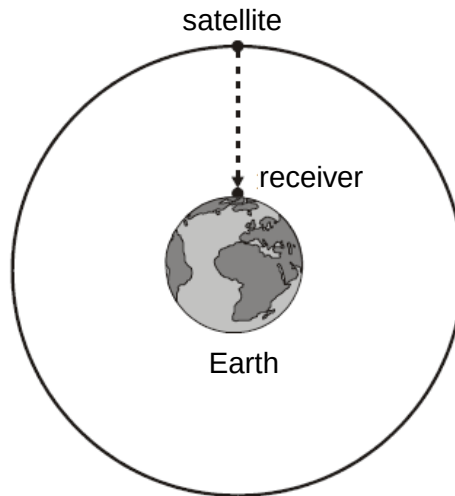
Which row is correct?

[1 mark]

	Orbital period	Total energy	
A	$T_X < T_Y$	$E_X < E_Y$	☐
B	$T_X < T_Y$	$E_Y < E_X$	☐
C	$T_Y < T_X$	$E_X < E_Y$	☐
D	$T_Y < T_X$	$E_Y < E_X$	☐

0 1

The Global Positioning System (GPS) is a system of satellites that transmit radio signals which can be used to locate the position of a receiver anywhere on Earth.



A receiver at sea level detects a signal from a satellite in a circular orbit when it is passing directly overhead as shown in the diagram above. The satellite's orbit is 2.04×10^7 m above the Earth's surface.

0 1 . 1

Show that the gravitational field strength of the Earth at the position of the satellite is 0.56 N kg^{-1} .

mass of the Earth	=	$6.0 \times 10^{24} \text{ kg}$
mean radius of the Earth	=	6400 km

[3 marks]

0 1 . 2

Calculate the speed of the satellite in this orbit.

[2 marks]

speed = _____ m s^{-1}

01 . 3

Calculate the time period of the satellite in this orbit.

[2 marks]

time period = _____ s

Turn over for the next question

